

An Energy Based Motif Finding Algorithm Using Chromatin Immunoprecipitation on Microarray Data

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Abstract

Finding motifs is an important challenge in computational biology. This paper describes an improved Energy-Based Motif Finding (EBMF) algorithm employing the chromatin immunoprecipitation on microarray (ChIP-on-Chip) experiments data. Unlike most other computational methods for motif discovery, based on the strong-signal model, the EBMF algorithm can utilize additional weak-signal information from the ChIP-on-Chip data. Thus it shows successful performance even in cases other popular methods fail. Our goal is to improve the currently available EBMF algorithm by incorporating association information between positions in a motif and by allowing the length of target motif not to be fixed. Up to now, we have implement basic features of the EBMF and will continue to incorporate other features.

1 INTRODUCTION

To understand the regulation of gene expression, the process by which a DNA segment is decoded to form a protein, is one of the greatest challenges in the current molecular biology. Transcription and translation are two main steps for gene expression. During the transcription process, mRNA molecule is created by copying a gene segment from the DNA. In the translation process, the mRNA is decoded to produce a protein.

To initiate the transcription process for a gene, one or more corresponding proteins, called transcription factors, should bind to specific regions, usually called binding sites, in the promoter region of the gene. Generally, it is known that a transcription factor can bind to multiple different binding sites, but these sites usually have similar DNA sequence pattern and length. For most transcription factors,

the common patterns for their binding sites, called motifs, are still unknown.

Most of the currently available motif finding algorithms[1, 3, 5] are extremely based on the strong-signal model in which only strong-signal sequences (i.e. those that are already known to contain binding sites very similar to the motif) are considered as input and weak-signal sequences (i.e. those do not contain any sub-string very similar to the motif) are disregarded. Buhler and Tompa have studied the limitations of methods based on the strong-signal model[3] and Francis *et al* have further characterized the types of problematic instances that cannot be solved by most of the motif finding methods by introducing the notion of minimal number of binding sites rather than minimal number of strong signal sequences.[4]

A recent trend in motif-finding is to make use of computational methods based on microarray data allowing the genome-wide location analysis. The method combines a modified chromatin immunoprecipitation (ChIP) procedure, which has been previously used to study protein-DNA interactions at a small number of specific DNA sites, with DNA microarray analysis.[2, 4, 7] This so-called ChIP-on-Chip technology can give us information of how strongly a given transcriptional factor binds to a specific genomic region as well as where it binds to.

Francis *et al.* have introduced an energy-based motif finding model to capture previously-ignored weak-signal information. They used additional information from ChIP-on-Chip experiments and considered the binding strength (as measured experimentally by observing color intensity) of each available sequence. It can be easily assumed that the binding strength should relate to the degree of similarity between the motif and the binding sites in each sequence.

Based on the binding strength, their model considered the amount of signals that a sequence actually contains. By comparison with the results of other famous motif finding programs such as AlignACE and MEME, they have shown the high performance of their Energy-Based Motif Finding (EBMF) algorithm. However, their algorithm does not consider any association between positions in each motif and the length of the expected motif should be fixed at the first step.

Our ultimate goal of this project is to develop an improved EBMF algorithm that can utilize association information between positions in a motif, whose length may not be known. For this project, we have two distinctive phases; the first is to implement the basic features of EBMF and the second is to improve the algorithm equipped with the features

described above.

2 METHODOLOGY

2.1 Energy-Based Model

The following energy-based model is based primarily on the Francis *et al.*' previous work[4]. In the microarray experiment, some copies of transcription factor (TF) may bind to the binding sites of different sequences s_i in the genome while some copies of the transcription factor and sequences are free. Let e_i be the average binding energy between the transcription factor and sequence s_i . Then $e_i = \ln(K_e)$ where the binding constant $K_e = [TF \bullet s_i] / [TF][s_i]$ (number of sequences bound to the transcription factors over number of free sequences) with the binding reaction modeled by $TF + s_i \leftrightarrow TF \bullet s_i$. For each sequence, we have a color intensity which represents the value of K_e . Thus, the color intensity can be used to estimate the binding energy between the transcription factor and each input sequence.

Let sequence s_i contain B_i binding sites and $E(b_{ij}, M)$ be the binding energy between the transcription factor and the j -th binding site b_{ij} in sequence s_i . The probability that the transcription factor binds to b_{ij} is

$$P_{ij} = \frac{e^{-E(b_{ij}, M)}}{\sum_{k=1}^{B_i} e^{-E(b_{ik}, M)}} \quad (1)$$

and we define $E_{total}(s_i, M)$ to be the expected binding energy between the TF and sequence s_i when there are at least one binding site in s_i that is bound by the TF.

Now, we want to find a $4 \times l$ energy matrix M to minimize the following *prediction error*, where l is the length of binding sites and t sequences $S = \{s_i\}$ in which each sequence s_i has a corresponding

binding energy e_i

$$\sum_{i=1}^t (E_{total}(s_i, M) - e_i)^2 \quad (2)$$

The initial energy matrices are refined to minimize the prediction error in EM-like iteration. Until there is no further decrease in the prediction error or the number of iterations reaches a given value, the iteration process is repeated.

2.2 Solving Linear Equations

We formulate an equation by setting expected binding energy to the given binding energy of that sequence, that is, $E_{total}(s_i, M') = e_i$. We only need $4 \times l$ distinctive equations to obtain a new energy matrix. In addition to the QR decomposition method[8] that was employed by Francis *et al.*, we also implemented Downhill Simplex Method[6, 8] to solve the linear equations since the matrices to be decomposed are singular up to the computational precision in some cases.

To create the linear equations, we convert each character in b_{ij} for any j in BS_{ik} to a 4-dimensional vector by using (1,0,0,0), (0,1,0,0), (0,0,1,0), and (0,0,0,1) for "A", "c", "G", and "T" respectively. The resulting $4l$ -dimensional vector v_{ij} is used to represent the binding site b_{ij} of length l . Then, the equations are represented as follows,

$$\left\{ \sum_{j=1}^{B_i} \left[\frac{p_{ij}}{1 - \prod_{BS_{ik}} \left(1 - \sum_{b_{im} \in BS_{ik}} P_{im} \right)} \right] \times v_{ij} \right\} \times V(M')^T = e_i \quad (3)$$

Here BS_{ik} means the divided groups of B_i binding sites of b_{ij} . For any two binding sites in the same group BS_{ik} , the distance between them is within d_{min} while the distance between any two binding sites in different groups is larger than d_{min} .

3 IMPLEMENTATION AND RESULTS

The first phase of this project that aims at implementing the basic features of EBMF has been undertaken. We have implemented EBMF in Perl and the program can perform iterative searches of candidate motif matrix from a given initial matrix. A test run on yeast Gal4 motif data[2] showed that our current program can predict proper energy matrix once we have proper initial energy matrix. However, some parts of the program such as generating initial energy matrix based on the given experimental data are not fully functional yet.

4 DISCUSSION

In this paper, we have introduced an energy-based motif finding algorithm based on Francis *et al.*'s previous work. Unlike most of the currently available motif finding algorithms, it can utilize weak-signal information from ChIP-on-Chip experiment data, thus greatly improve the performance of overall motif finding process. Our primary goal up to now has been implementing the basic EBMF algorithm in our own program.

Though our program is not fully functional yet, it outperforms Francis *et al.*' program in the aspect that ours can handle singular matrices which are very commonly observed in the EBMF. Thus it gives more accurate results.

The ultimate goal of this project is to develop an improved EBMF algorithm that can utilize association information between positions in a motif, whose length may not be known. Our future works include the followings.

- Implementation of either the Powell's method[8] or the conjugate gradient method[8] for faster linear equation solution than the currently im-

plemented downhill simplex method.

- Improvement of the initial energy matrix generation that allows various lengths of target motifs.
- Application of association information between positions in a motif by using 1st or higher order Markov chains.

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